

Derivation of $\mathbb{E}[p]$

Definitions

1. Vectors. Fix an integer $d \geq 2$. Vectors live in $\mathbb{R}^d$ and are written $v = (v_1, \dots, v_d)$. The dot product is $v \cdot w = \sum_i v_i w_i$ and the norm is $\|v\| = \sqrt{v \cdot v}$.

2. The sphere and its area. The unit sphere is $S^{d-1} = \{v \in \mathbb{R}^d : \|v\| = 1\}$, a $(d-1)$ -dimensional surface. Write dA_{d-1} for its surface-area element and

$$A_{d-1} = \int_{S^{d-1}} dA_{d-1}$$

for the total area. The same objects one dimension down — $S^{d-2} \subset \mathbb{R}^{d-1}$, its area element dA_{d-2} , and its total area A_{d-2} — are defined identically.

3. Uniform direction and expectation. Let u be a random vector drawn uniformly on S^{d-1} : for any region $R \subset S^{d-1}$, $\Pr(u \in R) = A(R)/A_{d-1}$, where $A(R)$ is the surface area of R . For a function g on the sphere, its expectation is

$$\mathbb{E}[g(u)] = \frac{1}{A_{d-1}} \int_{S^{d-1}} g(u) dA_{d-1}.$$

4. The target. For unit vectors t and u , let θ be their angle, so $\cos \theta = t \cdot u$. Define the harm

$$p = \max(0, -\cos \theta).$$

We compute $\mathbb{E}[p]$ with t fixed and u uniform (Definition 3).

5. Slice coordinates. For u with $|u_1| < 1$, set the height

$$c = u_1, \quad -1 < c < 1,$$

and the tail direction

$$\omega = \frac{(u_2, \dots, u_d)}{\sqrt{1 - c^2}} \in S^{d-2}.$$

The reconstruction map $\Phi : (-1, 1) \times S^{d-2} \rightarrow S^{d-1}$ is

$$\Phi(c, \omega) = \left(c, \sqrt{1 - c^2} \omega \right),$$

which inverts $u \mapsto (c, \omega)$; it is a bijection onto the sphere minus the two poles $u_1 = \pm 1$.

6. Induced volume form (Gram determinant). Let Ψ parametrize a k -dimensional surface, with tangent vectors $\tau_1, \dots, \tau_k$. The Gram matrix is $G = (g_{ij})$ with $g_{ij} = \tau_i \cdot \tau_j$. Its determinant is the squared k -volume of the parallelepiped spanned by the tangents, so the induced surface element is

$$dA = \sqrt{\det G} \cdot (d \text{ of parameter measure}).$$

For Φ in Definition 5 the tangents are $\partial\Phi/\partial c$ and the tangent vectors of S^{d-2} , and the parameter measure is $dc dA_{d-2}$.

7. Gamma and Beta functions. For $z > 0$,

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx \quad (\text{so } \Gamma(n) = (n-1)! \text{ and } \Gamma(\tfrac{1}{2}) = \sqrt{\pi}).$$

For $p, q > 0$,

$$B(p, q) = \int_0^1 x^{p-1} (1-x)^{q-1} dx = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}.$$

Derivation

Step 1: reduction. The uniform measure (Definition 3) is rotation-invariant, and p depends only on the angle between t and u , so we may rotate coordinates to place $t = e_1 = (1, 0, \dots, 0)$. Then $\cos \theta = t \cdot u = u_1 = c$ (Definition 5), so

$$\mathbb{E}[p] = \mathbb{E}[\max(0, -c)].$$

The map $u \mapsto -u$ preserves the uniform measure and sends c to $-c$, so c has a symmetric distribution and

$$\mathbb{E}[\max(0, -c)] = \tfrac{1}{2} \mathbb{E}|c|.$$

(Indeed $\mathbb{E}|c| = \mathbb{E}[\max(0, c)] + \mathbb{E}[\max(0, -c)]$, and symmetry makes the two terms equal.) It remains to compute $\mathbb{E}|c|$.

Step 2: the induced measure. Apply Definition 6 to Φ . The c -tangent is

$$\frac{\partial\Phi}{\partial c} = \left(1, -\frac{c}{\sqrt{1-c^2}}\omega\right), \quad \left\|\frac{\partial\Phi}{\partial c}\right\|^2 = 1 + \frac{c^2}{1-c^2} = \frac{1}{1-c^2}.$$

A tangent vector v to S^{d-2} (so $v \cdot \omega = 0$) pushes forward to $(0, \sqrt{1-c^2}v)$, which is orthogonal to $\partial\Phi/\partial c$ and scaled by $\sqrt{1-c^2}$ in each of the $d-2$ tangential directions. The Gram matrix is therefore block-diagonal, and

$$\sqrt{\det G} = \frac{1}{\sqrt{1-c^2}} \cdot (1-c^2)^{(d-2)/2} = (1-c^2)^{(d-3)/2}.$$

Thus

$$dA_{d-1} = (1-c^2)^{(d-3)/2} dc dA_{d-2}.$$

Step 3: the marginal density of c . The band $\{c \leq u_1 \leq c + dc\}$ has area

$$A_{d-2} (1-c^2)^{(d-3)/2} dc,$$

so by Definition 3 the density of c is

$$f(c) = \frac{A_{d-2}}{A_{d-1}} (1 - c^2)^{(d-3)/2}, \quad -1 \leq c \leq 1.$$

Step 4: the areas. Integrating the measure of Step 2 over all c :

$$A_{d-1} = A_{d-2} \int_{-1}^1 (1 - c^2)^{(d-3)/2} dc = A_{d-2} B\left(\frac{1}{2}, \frac{d-1}{2}\right) = A_{d-2} \frac{\sqrt{\pi} \Gamma(\frac{d-1}{2})}{\Gamma(\frac{d}{2})}$$

(Definitions 7; the substitution $x = c^2$ turns the integral into B). With the base cases $A_0 = 2$ and $A_1 = 2\pi$, this recursion solves by induction to

$$A_{d-1} = \frac{2\pi^{d/2}}{\Gamma(d/2)} \quad (\text{check: } d = 2 \text{ gives } 2\pi, \ d = 3 \text{ gives } 4\pi).$$

Therefore the ratio in $f(c)$ is

$$\frac{A_{d-2}}{A_{d-1}} = \frac{\Gamma(d/2)}{\sqrt{\pi} \Gamma((d-1)/2)}.$$

Step 5: the average of $|c|$. From Step 3,

$$\mathbb{E} |c| = \int_{-1}^1 |c| f(c) dc = \frac{A_{d-2}}{A_{d-1}} \int_{-1}^1 |c| (1 - c^2)^{(d-3)/2} dc.$$

The integrand is even, so restrict to $[0, 1]$ and double, then substitute $w = c^2$ ($dw = 2c dc$):

$$\int_{-1}^1 |c| (1 - c^2)^{(d-3)/2} dc = 2 \int_0^1 c (1 - c^2)^{(d-3)/2} dc = \int_0^1 (1 - w)^{(d-3)/2} dw = \frac{2}{d-1}.$$

Step 6: assemble.

$$\mathbb{E}[p] = \frac{1}{2} \mathbb{E} |c| = \frac{1}{2} \cdot \frac{A_{d-2}}{A_{d-1}} \cdot \frac{2}{d-1} = \frac{\Gamma(d/2)}{\sqrt{\pi} (d-1) \Gamma((d-1)/2)}.$$

Step 7: large d . Stirling's expansion for the gamma ratio is

$$\frac{\Gamma(x + \frac{1}{2})}{\Gamma(x)} = \sqrt{x} \left(1 - \frac{1}{8x} + \frac{1}{128x^2} + O(x^{-3}) \right),$$

where $O(x^{-3})$ denotes a remainder bounded by a constant times x^{-3} . With $x = \frac{d-1}{2}$,

$$\mathbb{E}[p] = \frac{1}{\sqrt{\pi}(d-1)} \frac{\Gamma(d/2)}{\Gamma((d-1)/2)} = \frac{1}{\sqrt{2\pi}(d-1)} \left(1 - \frac{1}{4(d-1)} + O(d^{-2}) \right).$$

Expanding $(d-1)^{-1/2} = d^{-1/2}(1 + \frac{1}{2d} + O(d^{-2}))$,

$$\mathbb{E}[p] = \frac{1}{\sqrt{2\pi d}} \left(1 + \frac{1}{4d} + O(d^{-2}) \right) \approx \frac{1}{\sqrt{2\pi d}}.$$